Fundamentos de Python / Python Fundamentals

Tema 3 / Topic 3: Integración y EDO con SymPy
Integration and ODEs with SymPy

May 3, 2026

Cálculo para la Computación / Calculus for Computing

Índice / Outline

Primitivas / Primitives

Primitivas / Primitives

ES Calculemos la primitiva de la función.

EN Let us compute the primitive of the function.

$$f(x) = ax^2 + bx + c$$

```python
from sympy import integrate
from sympy.abc import a, b, c, x

f = a*x**2 + b*x + c
primitive = integrate(f, x)
print(primitive)
# Output: a*x**3/3 + b*x**2/2 + c*x
```

ES La primitiva general es / EN: The general primitive is:

$$\frac{ax^3}{3} + \frac{bx^2}{2} + cx + C$$

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ES Verificar las siguientes primitivas.

EN Verify the following primitives.

1. $$\int \cos(\ln x)\,dx = \frac{x\sin(\ln x)}{2} + \frac{x\cos(\ln x)}{2} + C$$
2. $$\int (x+5)\cos x\,dx = x\sin x + 5\sin x + \cos x + C$$
3. $$\int \frac{\cos(2x)}{\sin^3(2x)}\,dx = -\frac{1}{4\sin^2(2x)} + C$$
4. $$\int \frac{1+x}{1+\sqrt{x}}\,dx = \frac{2x^{3/2}}{3} + 4\sqrt{x} - 1 - 4\ln(\sqrt{x}+1) + C$$

Verificando una integral / Verifying an Integral

ES Ejemplo: verificar $\int \cos(\ln x)\, dx$.

EN Example: verify $\int \cos(\ln x)\, dx$.

```python
from sympy import integrate, cos, log, x, simplify

# Compute the integral
result = integrate(cos(log(x)), x)
print(simplify(result))

# Output: x*sin(log(x))/2 + x*cos(log(x))/2
```

Ecuaciones diferenciales / Differential Equations

EDO / ODE: ejemplo básico

ES Resolvamos la ecuación diferencial.

EN Let us solve the differential equation.

$$y' = 6x^2 - 3x^2 y$$

```python
import sympy as sp

x = sp.Symbol('x')
y = sp.Function('y')

# Define the differential equation
eq = sp.Eq(y(x).diff(x), 6*x**2 - 3*x**2*y(x))

# Solve the equation
solution = sp.dsolve(eq, y(x))
print(solution)
```

EDO / ODE: ejemplo básico

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# Solve the equation
solution = sp.dsolve(eq, y(x))
print(solution)
```

EDO: solución general / ODE: general solution

ES La solución general incluye una constante C_1.
EN The general solution includes a constant C_1.

ES Verifica sustituyendo en la ecuación.
EN Check it by substituting back into the equation.

Ejercicio EDO 1 / ODE Exercise 1

ES Verificar que la siguiente función resuelve la EDO.
EN Verify that the following function solves the ODE.

$$(x^2 + 4)y' = xy, \qquad y = C_1\sqrt{x^2 + 4}.$$

Ejercicio EDO 2 / ODE Exercise 2

ES Verificar que la siguiente función resuelve la EDO.

EN Verify that the following function solves the ODE.

$$xy^2 + x + (yx^2)y' = 0, \qquad y = \pm\sqrt{\frac{C_1}{x^2} - 1}.$$

EDO: verificación en SymPy / ODE: SymPy check

ES Verificar $(x^2+4)y' = xy$.

EN Verify $(x^2+4)y' = xy$.

```python
import sympy as sp

x = sp.Symbol('x')
y = sp.Function('y')

# (x^2 + 4)y' - xy = 0
eq = sp.Eq((x**2 + 4)*y(x).diff(x), x*y(x))

solution = sp.dsolve(eq, y(x))
print(solution)
# Output: y(x) = C1*sqrt(x**2 + 4)
```

Integrales definidas y dobles

Integral definida / Definite Integral

ES Calcular la integral definida.

EN Compute the definite integral.

$$\int_0^{\pi/2} \cos(x)\, dx$$

```python
from sympy import integrate, cos, pi, x

result = integrate(cos(x), (x, 0, pi/2))
print(result)
# Output: 1
```

Integral definida / Definite Integral

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result = integrate(cos(x), (x, 0, pi/2))
print(result)
# Output: 1
```

Ejemplo: área de un círculo / Example: circle area

ES Área de un círculo de radio 2.
EN Area of a circle of radius 2.

Integral:

$$A = \int_{-2}^{2} 2\sqrt{4 - x^2}\, dx$$

Ecuación / Equation:

$$x^2 + y^2 = 4, \qquad y = \pm\sqrt{4 - x^2}.$$

Círculo en SymPy / Circle in SymPy

```python
from sympy import integrate, sqrt, x, pi

area = integrate(2*sqrt(4 - x**2), (x, -2, 2))
print(area)
# Output: 4*pi
```

ES Resultado esperado $A = 4\pi$.

EN Expected result $A = 4\pi$.

Integral doble / Double Integral

ES Área de un rectángulo.
EN Area of a rectangle.

$$\int_a^b \int_c^d 1 \, dy \, dx$$

```python
from sympy import integrate, simplify
from sympy.abc import x, y, a, b, c, d

I1 = integrate(1, (y, c, d))
result = simplify(integrate(I1, (x, a, b)))
print(result)
# Output: (b - a)*(d - c)
```

Integral doble / Double Integral

ES Área de un rectángulo.

EN Area of a rectangle.

$$\int_a^b \int_c^d 1\, dy\, dx$$

```
from sympy import integrate, simplify
from sympy.abc import x, y, a, b, c, d

I1 = integrate(1, (y, c, d))
result = simplify(integrate(I1, (x, a, b)))
print(result)
# Output: (b - a)*(d - c)
```

Círculo como integral doble / Circle as double integral

ES Área del círculo de radio 2.
EN Area of the circle of radius 2.

Cartesian / Cartesiano:

$$A = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 1 \, dy \, dx$$

Polar / Polares:

$$A = \int_{0}^{2\pi} \int_{0}^{2} r \, dr \, d\theta$$

Integral impropia típica / Typical improper integral

ES Comparar $1/x$ y $1/x^2$.
EN Compare $1/x$ and $1/x^2$.

$$\int_1^{+\infty} \frac{1}{x^2}\,dx \quad \text{(converge)}$$

Resumen / Summary

Resumen / Summary

- ES: `integrate()` para integrales simples y dobles. EN: `integrate()` for single and double integrals.
- ES: `dsolve()` para EDO. EN: `dsolve()` for ODEs.
- ES: Elegir coordenadas cartesianas o polares según la región. EN: Choose Cartesian or polar coordinates depending on the region.

Ejercicios 15 min / 15-minute Activities

Exercise 1 (15 min)

ES Objetivo: repasar integración y EDO separables.
EN Goal: review integration and separable ODEs.

1. ES: Con SymPy, comprobar que la integral vale π. EN: With SymPy, check that the integral equals π.

$$\int_0^{\pi} x \sin x \, dx = \pi$$

2. ES: Resolver con `dsolve` la EDO con condición inicial. EN: Solve with `dsolve` the ODE with initial condition.

$$y' = x\,y, \qquad y(0) = 2$$

Hints / Pistas (Exercise 1)

```python
from sympy import integrate, sin, pi, x, N

I = integrate(x*sin(x), (x, 0, pi))
print(I)
print(N(I))
```

```python
import sympy as sp

x = sp.Symbol('x')
y = sp.Function('y')

ode = sp.Eq(y(x).diff(x), x*y(x))
sol = sp.dsolve(ode, ics={y(0): 2})
print(sol)
```

Hints / Pistas (Exercise 1)

```python
from sympy import integrate, sin, pi, x, N

I = integrate(x*sin(x), (x, 0, pi))
print(I)
print(N(I))
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import sympy as sp

x = sp.Symbol('x')
y = sp.Function('y')

ode = sp.Eq(y(x).diff(x), x*y(x))
sol = sp.dsolve(ode, ics={y(0): 2})
print(sol)
```

Exercise 2 (15 min)

ES Objetivo: conectar área, integrales y gráficas.
EN Goal: connect area, integrals and plots.

1. ES: Graficar $f(x) = \sqrt{x}$ y $g(x) = x^2$ y sombrear la región.
   EN: Plot $f(x) = \sqrt{x}$ and $g(x) = x^2$ and shade the region.
2. ES/EN: Comprobar con SymPy que el área es $1/3$.

$$A = \int_0^1 (\sqrt{x} - x^2)\, dx = \frac{1}{3}$$

Hints / Pistas (Exercise 2)

```python
from sympy import symbols, sqrt, integrate

x = symbols('x')
f = sqrt(x)
g = x**2
A = integrate(f - g, (x, 0, 1))
print(A)
```

Python Basics – Topic 4

Sequences, Series and Taylor Expansions

Sucesiones, Series y Desarrollos de Taylor

24 de mayo de 2026

Calculus for Computing / Cálculo para Informática

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Sequences and Limits / Sucesiones y Límites

Computing Limits of Sequences / Calculando Límites de Sucesiones

SymPy can compute limits of sequences as $n \to \infty$ — SymPy puede calcular límites de sucesiones cuando $n \to \infty$

```
from sympy import symbols, Limit, oo, sin

# Define symbolic variable / Define variable
    simb lica
n = symbols('n', integer=True)

# Converging sequence / Sucesi n convergente
a_n = 1 / n
limit = Limit(a_n, n, oo)
print(limit.doit())  # Output: 0
```

```
# Oscillating sequence / Sucesi n oscilante
a_f = sin(n)
limit1 = Limit(a_f, n, oo)
print(limit1.doit())  # Output: AccumBounds(-1, 1)
```

Computing Limits of Sequences / Calculando Límites de Sucesiones

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limit = Limit(a_n, n, oo)
print(limit.doit())  # Output: 0
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# Oscillating sequence / Sucesi n oscilante
a_f = sin(n)
limit1 = Limit(a_f, n, oo)
print(limit1.doit())  # Output: AccumBounds(-1, 1)
```

Examples / Ejemplos

Calculate the following limits: / Calcula los siguientes límites:

1. $\lim_{n\to\infty} \dfrac{n^2+1}{2n^2-3}$
2. $\lim_{n\to\infty} \dfrac{n}{e^n}$
3. $\lim_{n\to\infty} \left(\sqrt{n^2+n}-n\right)$
4. $\lim_{n\to\infty} \dfrac{(-1)^n}{n}$
5. $\lim_{n\to\infty} \left(\dfrac{3n-7}{3n+5}\right)^{2n}$
6. $\lim_{n\to\infty} \sin \dfrac{\pi n-1}{2-3n}$
7. $\lim_{n\to\infty} \left(\sqrt[3]{n+1}-\sqrt[3]{n}\right)$
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Solution / Solución: $\lim_{n\to\infty} \frac{n^2+1}{2n^2-3}$

```python
from sympy import symbols, Limit, oo, simplify

n = symbols('n', integer=True)
a_n = (n**2 + 1) / (2*n**2 - 3)

limit = Limit(a_n, n, oo)
result = limit.doit()
print(f"Limit: {result}")
# Output: Limit: 1/2
```

Manual / A mano:

$$\frac{n^2+1}{2n^2-3} = \frac{1+1/n^2}{2-3/n^2} \xrightarrow{n\to\infty} \frac{1}{2}$$

Result / Resultado: The limit is $\frac{1}{2}$

Solution / Solución: $\lim_{n\to\infty} \dfrac{n^2+1}{2n^2-3}$

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Solution / Solución: $\lim_{n\to\infty} \frac{n^2+1}{2n^2-3}$

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Result / Resultado: The limit is $\frac{1}{2}$

Solution / Solución: $\lim_{n\to\infty}\left(\sqrt{n^2+n}-n\right)$

```python
from sympy import symbols, Limit, oo, sqrt, simplify

n = symbols('n', integer=True, positive=True)
a_n = sqrt(n**2 + n) - n

# Simplify first (rationalization) / Simplifica (
    racionaliza)
simplified = simplify(a_n)
print(f"Simplified: {simplified}")

limit = Limit(a_n, n, oo)
result = limit.doit()
print(f"Limit: {result}")
# Output: Limit: 1/2
```

Trick / Truco (rationalization / racionalización):

$$\left(\sqrt{n^2+n}-n\right)\cdot\frac{\sqrt{n^2+n}+n}{\sqrt{n^2+n}+n}=\frac{n}{\sqrt{n^2+n}+n}\xrightarrow{n\to\infty}\frac{1}{2}$$

Solution / Solución: $\lim_{n\to\infty}\left(\sqrt{n^2+n}-n\right)$

```python
from sympy import symbols, Limit, oo, sqrt, simplify

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Trick / Truco (rationalization / racionalización):

$$\left(\sqrt{n^2+n}-n\right)\cdot\frac{\sqrt{n^2+n}+n}{\sqrt{n^2+n}+n}=\frac{n}{\sqrt{n^2+n}+n}\xrightarrow{n\to\infty}\frac{1}{2}$$

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from sympy import symbols, Limit, oo, sqrt, simplify

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a_n = sqrt(n**2 + n) - n

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$$\left(\sqrt{n^2+n}-n\right)\cdot\frac{\sqrt{n^2+n}+n}{\sqrt{n^2+n}+n}=\frac{n}{\sqrt{n^2+n}+n}\xrightarrow{n\to\infty}\frac{1}{2}$$

Numerical Series / Series Numéricas

Numerical Series in SymPy / Series en SymPy

SymPy can compute infinite series using the Sum function — SymPy puede calcular series infinitas con la función Sum

```
from sympy import Sum, factorial, oo, symbols

n = symbols('n', integer=True)

# Converging series / Serie convergente: sum of 1/n!
sumn = Sum(1/factorial(n), (n, 0, oo))
print(f"Sum of 1/n!: {sumn.doit()}")
# Output: E  (Euler's number / n mero de Euler)
```

```
# Diverging series / Serie divergente: harmonic series
    / arm nica
summ = Sum(1/n, (n, 1, oo))
print(f"Harmonic series: {summ.doit()}")
# Output: oo (diverges / diverge)
```

SymPy can compute infinite series using the `Sum` function — SymPy puede calcular series infinitas con la función `Sum`

```
from sympy import Sum, factorial, oo, symbols

n = symbols('n', integer=True)

# Converging series / Serie convergente: sum of 1/n!
sumn = Sum(1/factorial(n), (n, 0, oo))
print(f"Sum of 1/n!: {sumn.doit()}")
# Output: E  (Euler's number / n mero de Euler)
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```
# Diverging series / Serie divergente: harmonic series
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summ = Sum(1/n, (n, 1, oo))
print(f"Harmonic series: {summ.doit()}")
# Output: oo (diverges / diverge)
```

Numerical Series in SymPy / Series en SymPy

SymPy can compute infinite series using the Sum function —
SymPy puede calcular series infinitas con la función Sum

```
from sympy import Sum, factorial, oo, symbols

n = symbols('n', integer=True)

# Converging series / Serie convergente: sum of 1/n!
sumn = Sum(1/factorial(n), (n, 0, oo))
print(f"Sum of 1/n!: {sumn.doit()}")
# Output: E  (Euler's number / n mero de Euler)
```

```
# Diverging series / Serie divergente: harmonic series
    / arm nica
summ = Sum(1/n, (n, 1, oo))
print(f"Harmonic series: {summ.doit()}")
# Output: oo (diverges / diverge)
```

Pretty Printing / Impresión Matemática

For better mathematical output: / Para mejor visualización matemática:

```python
import sympy as sp

# Initialize pretty printing / Inicializa impresi n matem tica
sp.init_printing(use_latex='mathjax')

n = sp.symbols('n', integer=True)
sumn = sp.Sum(1/sp.factorial(n), (n, 0, sp.oo))

display(sumn)        # Symbolic / Simb lico
display(sumn.doit()) # Evaluated / Evaluado
```

Note / Nota: Useful in Jupyter notebooks for LaTeX-rendered output — Útil en Jupyter para salida renderizada en LaTeX

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Examples / Ejemplos

Calculate the following series: / Calcula las siguientes series:

1. $\displaystyle\sum_{n=0}^{\infty}\left(\frac{2}{3}\right)^n$
2. $\displaystyle\sum_{n=0}^{\infty}\left(\frac{3}{2}\right)^n$
3. $\displaystyle\sum_{n=0}^{\infty}\frac{i^n}{3^n}$
4. $\displaystyle\sum_{n=0}^{\infty}\frac{2^n}{n!}$
5. $\displaystyle\sum_{n=0}^{\infty}\frac{n+3}{2^n}$
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Geometric Series / Serie Geométrica: $\sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n$

```python
from sympy import Sum, symbols, oo, Rational

n = symbols('n', integer=True)

# Geometric series r = 2/3 / Serie geom trica r = 2/3
series = Sum(Rational(2,3)**n, (n, 0, oo))
result = series.doit()

print(f"Sum: {result}")
# Output: Sum: 3
```

Formula / Fórmula: For / Para $|r| < 1$:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} = \frac{1}{1-\frac{2}{3}} = 3$$

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Exponential Series / Serie Exponencial: $\sum_{n=0}^{\infty} \frac{2^n}{n!}$

```python
from sympy import Sum, factorial, symbols, oo, exp

n = symbols('n', integer=True)

# This is the Taylor series for e^2 / Serie de Taylor de e^2
series = Sum(2**n / factorial(n), (n, 0, oo))
result = series.doit()

print(f"Sum: {result}")            # Output: exp(2)
print(f"Numerical: {float(result):.6f}")  # 7.389056
```

Recognition / Reconocimiento: This is e^2, from the Taylor series of e^x — Es e^2, de la serie de Taylor de e^x

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Ratio Test (D'Alembert) / Criterio del Cociente

Automate the ratio $\ell = \lim \frac{a_{n+1}}{a_n}$ with SymPy — Automatiza el cociente con SymPy

```
from sympy import symbols, Limit, oo, factorial,
    simplify

n = symbols('n', integer=True, positive=True)

# Converging / Convergente: sum 2^n / n!
a_n = 2**n / factorial(n)
ratio = simplify(a_n.subs(n, n+1) / a_n)
L = Limit(ratio, n, oo).doit()
print(f"L={L}")    # L=0 < 1 => converges / converge
```

```
# Diverging / Divergente: sum n^n / n!
a_n2 = n**n / factorial(n)
ratio2 = simplify(a_n2.subs(n, n+1) / a_n2)
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print(f"L={L2}")  # L=E > 1 => diverges / diverge
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Exercises: Ratio Test / Ejercicios: Criterio del Cociente

Determine convergence: / Determina la convergencia:

1. $\displaystyle\sum_{n=1}^{\infty} \frac{3^n}{n!}$
2. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n \cdot n^2}{n!}$
3. $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{n^n}$
4. $\displaystyle\sum_{n=1}^{\infty} \frac{2^{n-2}}{n+1}$
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Remember / Recuerda: if / si $\ell = 1$, the test is inconclusive / el criterio es inconclusivo

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Taylor Series / Series de Taylor

Taylor Series Expansion / Desarrollo en Serie de Taylor

Compute Taylor/Maclaurin series expansions — Calcula desarrollos en serie de Taylor/Maclaurin

```
from sympy import symbols, exp, series

x = symbols('x')

# Taylor series of e^x around x=0 (Maclaurin)
# Serie de Taylor de e^x en x=0 (Maclaurin)
taylor_exp = series(exp(x), x, 0, 6)

print(taylor_exp)
# 1 + x + x**2/2 + x**3/6 + x**4/24
# + x**5/120 + O(x**6)

# Remove Big-O term / Quitar t rmino resto
poly = taylor_exp.removeO()
```


Note / Nota: O(x**6) is the remainder / es el residuo

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Examples / Ejemplos

Calculate the Taylor series (around $x = 0$) for: / Calcula la serie de Taylor (en $x = 0$) de:

1. $\log(1 + x)$
2. $(1 + x)^\alpha$ (binomial series / serie binomial)
3. $\sin x$
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Example / Ejemplo: Taylor Series of $\sin x$ / Serie de Taylor de $\sin x$

```python
from sympy import symbols, sin, series

x = symbols('x')

# Taylor series of sin(x) up to degree 10
# Serie de Taylor de sin(x) hasta grado 10
taylor_sin = series(sin(x), x, 0, 11)

print(taylor_sin)
# x - x**3/6 + x**5/120 - x**7/5040
# + x**9/362880 + O(x**11)
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Pattern / Patrón: Only odd powers, alternating signs! / ¡Solo potencias impares, signos alternos!

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$$\sin x = x - x^3 + x^5 - x^7 + \cdots = \sum^{\infty} (-1)^n x^{2n+1}$$

Example / Ejemplo: Taylor Series of $\log(1+x)$

```python
from sympy import symbols, log, series

x = symbols('x')

# Taylor series of log(1+x)
# Serie de Taylor de log(1+x)
taylor_log = series(log(1+x), x, 0, 8)

print(taylor_log)
# x - x**2/2 + x**3/3 - x**4/4
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Series / Serie:

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

Convergence / Convergencia: $|x| \leq 1,\ x \neq -1$

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Convergence / Convergencia: $|x| \leq 1,\ x \neq -1$

Example / Ejemplo: Binomial Series $(1+x)^\alpha$ / Serie Binomial

```python
from sympy import symbols, series

x = symbols('x')
alpha = symbols('alpha')

# Binomial series / Serie binomial
binomial = (1+x)**alpha
taylor_binom = series(binomial, x, 0, 5)

print(taylor_binom)
# 1 + alpha*x
# + alpha*(alpha-1)*x**2/2
# + alpha*(alpha-1)*(alpha-2)*x**3/6
# + O(x**4)
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$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \cdots$$

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$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \cdots$$

Summary / Resumen

Key Takeaways / Conceptos Clave

1. **Limits / Límites:** Use `Limit(expr, n, oo).doit()` for sequence limits / para límites de sucesiones
2. **Series:** Use `Sum(expr, (n, 0, oo)).doit()` to compute infinite series / para calcular series infinitas
3. **Taylor Series / Series de Taylor:** Use `series(f, x, x0, n)` for Taylor expansions / para desarrollos de Taylor
4. **Simplification / Simplificación:** Always consider simplifying expressions before computing limits / Simplifica siempre antes de calcular límites

Practice / Practica: Work through the examples to master these techniques! / ¡Resuelve los ejemplos para dominar estas técnicas!

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